Roshan Karathan Pradeep

MECHANICAL DESIGN ENGINEER

Stockholm, Sweden | +46 769740529 | rpradeep@kth.se | linkedin.com/in/roshanaerodynamics

Summary

Mechanical design engineer and KTH master's student in Vehicle Engineering, with experience taking mechanical systems from concept through detailed design, manufacturing drawings, and physical validation. Uses computer-aided design (CAD) and computational fluid dynamics (CFD) to solve thermal, structural, and 3D packaging problems, backed by hands-on prototype, laboratory, and workshop experience. Looking for a mechanical design role in the automotive industry.

Experience

Research Assistant (Amanuens) — KTH Royal Institute of Technology *01/2026 - 08/2026*

- Led detail design and parametric 3D CAD modeling of suspension and primary structural components — subframe, A-arms, and aluminium suspension parts — for RCV-Dynamic, a four-wheel in-wheel-motor research vehicle (4 x 700 Nm) with individual steering, camber, drive/brake, and active suspension control, and owned release of the associated design documentation and manufacturing specifications.
- Produced manufacturing and assembly drawings with full geometric dimensioning and tolerancing (GD&T), setting datum schemes and tolerance callouts against each supplier's process capability to cut lead time.
- Sourced and onboarded new suppliers with wire electrical discharge machining (EDM) capability, applying design for manufacturing (DFM) changes to align part geometry with those processes.
- Selected and integrated commercial off-the-shelf (COTS) hardware including linear actuators, pedals, and radiators, packaging each unit and routing its wiring so harnesses cleared assembly paths and were protected against damage in rough use.
- Built and assembled prototype hardware in the laboratory, directed junior engineers through build and fit-up, carried out dimensional self-inspection against drawing, and contributed to peer design reviews.

R&D Engineer and Product Owner — Morphle Labs (YC20) *05/2024 - 05/2025*

- Designed precision optomechanical assemblies from concept through quality-control (QC) sign-off, including a custom linear roller bearing stage holding 1-micron tolerances for 100x high-magnification medical imaging devices.
- Served as product owner for the flagship MorphoLens 6, the MorphoLens 1, and bespoke machine variants, including remote and on-site field support.
- Ran design verification and validation on prototype builds and pilot machines, root-caused recurring failure modes, and codified the findings into standard design for manufacturing (DFM) and assembly guidelines.
- Resolved recurring field cable failures found during after-sales support: selected IGUS cable stock, designed a stress relief for each cable type, and defined mandatory routing paths and minimum bend radii for the MorphoLens 6, cutting electrical failures from 15 to 3 per month.
- Specified and integrated commercial off-the-shelf (COTS) components — linear roller bearings, rail blocks, special fasteners, cameras, and Raspberry Pi single-board computers — into precision assemblies, resolving mechanical fit, mounting, and cable routing against the surrounding structure.
- Coordinated with domestic and international part vendors and with purchasing to close manufacturability issues, specify heat treatment for hardened gears and shafts, and hold delivery schedules.

CFD Engineer Intern — Reude Technologies *01/2024 - 05/2024*

- Ran parametric external-aerodynamics CFD studies on unmanned aerial vehicle (UAV) wing geometries to support detailed design.
- Applied NSGA-II multi-objective optimization across an aerodynamic attachment design space, trading drag against aerodynamic efficiency to select the final configuration.

CFD Intern, Thermal and Flow Dynamics — Ashok Leyland *11/2023 - 04/2024*

- Optimized coolant flow distribution and reduced pressure drop in Ansys Fluent using design of experiments (DOE) and response surface methodology (RSM), reaching a flow uniformity index of 97%.
- Automated parametric geometry generation for an electric vehicle (EV) battery cold plate in Ansys SpaceClaim, enabling rapid DOE sweeps of flow and thermal performance.

Team Captain and Lead Design Engineer — Inferno Racing (SAE Baja) *2021 - 2024*

- Led a cross-functional team of 25 student engineers through the full design, build, and test cycle for the SAE Baja all-terrain vehicle competition, placing 3rd nationwide.
- Designed the transmission and suspension systems and validated the chassis against static and dynamic load cases by finite element analysis (FEA), achieving a minimum factor of safety of 1.5.
- Led vehicle manufacturing from raw stock through final assembly, specifying heat treatment for hardened gears and shafts and resolving process issues on the shop floor.

Workshop Intern — Mercedes-Benz *05/2022 - 06/2022*

- Diagnosed and rectified mechanical and electronic faults on customer vehicles in an authorized service workshop under short turnaround times.
- Operated Mercedes-Benz XENTRY Diagnosis for guided fault tracing, control-unit diagnostics, and scheduled maintenance.

Education

Master of Science (MSc), Vehicle Engineering (Road Track) *2025 - 2027*
KTH Royal Institute of Technology, Stockholm, Sweden

Bachelor of Engineering (BE), Mechanical Engineering *2024*
Anna University, India

- Overall grade: 8.7 / 10.
- Electives: Gas Dynamics (A+), Finite Element Analysis (A+), Heat and Mass Transfer (A+).

Skills

- **CAD and Drafting:** SolidWorks (advanced), Autodesk Fusion (advanced), geometric dimensioning and tolerancing (GD&T) (advanced), CATIA (intermediate), manufacturing and assembly drawings.
- **Design and Manufacturing:** Part and assembly design, commercial off-the-shelf (COTS) component and fastener selection and integration, packaging and cable routing, heat treatment specification, design for manufacturing (DFM), design verification and validation, prototype build and test, supplier sourcing and coordination.
- **Simulation and Analysis:** Ansys Workbench (intermediate), Ansys Fluent, Ansys SpaceClaim, OpenFOAM (intermediate), Adams Car, flow and cooling simulation, finite element method and analysis (FEM/FEA), design of experiments (DOE) and response surface methodology, parametric optimization.
- **Programming:** Python, MATLAB, Simulink.
- **Languages:** English (fluent), Swedish (basic, studying at SFI).

Projects

- Modeling and validation of the suspension system of a Volvo S90 in Adams Car.
  - Modeled the full suspension system in Adams Car to test, validate, and modify the vehicle suspension parameters, achieving a 5-second improvement in the standard slalom test.
- Eigenfrequency analysis of an aileron from an Airbus A321.
  - Performed vibration analysis on the carbon fiber reinforced polymer (CFRP) part to find its eigenfrequencies and to propose design changes that remove modes below 50 Hz.
- Simulink modeling of an electric wrench.
  - Modeled the nutrunner from scratch, building and tuning the controllers to correlate the model against the physical part.
- Performance analysis of a wing with and without aerodynamic attachments using CFD.
  - Tested a standard NACA 2412 airfoil with attachments including gurney flaps and serrated edges to assess their effect and where each is worth applying.